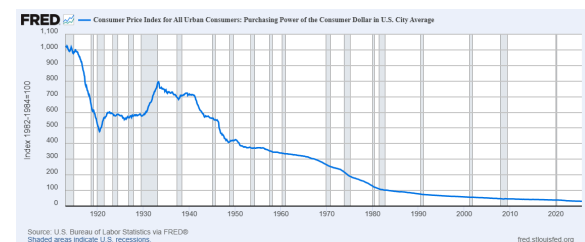


Competition in an Empty Room: Algorithmic Price Fixing

I. Introduction

In *The Wealth of Nations*,¹ Adam Smith used the metaphor of ‘an invisible hand’ to describe the unintended social benefits of self-interested merchants under certain conditions. Through this metaphor, Smith proposed that self-interested pursuit of profit can unintentionally incentivize efficiency and serve consumers. Modern corporations² often invoke his metaphor to suggest that markets can regulate themselves, yet economists continually debate when external regulation is necessary.

In neoclassical economic theory under perfectly competitive conditions, product prices adjust so that firms earn zero economic profit in the long run, and market supply and demand converge at a long-run equilibrium price.³ Sustained deviation, with little return to equilibrium prices or consistent economic inefficiency, is referred to as ‘market failure’; such failure is often a result of poor resource allocation, overwhelmingly dominant market share, or externalities.



At time of writing, customers are facing decreasing buying power and stagnant wages.⁴ The Federal Trade Commission, or FTC, has recently recognized algorithmic price-fixing as a

potential cause of these modern indicators of market inefficiency.⁵ Algorithmic price-fixing is a form of automated collusion that uses large anonymized consumer datasets to maximize profits for firms. This innovation has disrupted attempts at market regulation by obscuring firm accountability. While price-collusion itself is illegal, algorithms cannot be held personally responsible.⁶ FTC regulatory efforts, therefore, are limited by anonymized algorithmic shields from corporate accountability.

Price controls are not new; historians have cited the third-century Roman Emperor Diocletian⁷ as an example of attempting to control inflation by setting price ceilings for Roman goods. Despite its storied history, the price controls remain a challenge for modern regulators. Establishing modern regulatory solutions requires understanding the mechanics of price-fixing.

II. A Troubled Triumvirate

A. The Sherman Antitrust Act of 1890

In the United States, antitrust legislation began with the Sherman Antitrust Act of 1890.⁸ The law declared that, “[e]very contract, combination in the form of trust or otherwise, or conspiracy, in restraint of trade or commerce among the several States, or with foreign nations, is declared to be illegal.” Senator John Sherman of Ohio introduced the law to regulate monopolies and trusts that held dominant market share.⁹ Sherman and like-minded politicians recognized the future implications of corporate consolidation and instituted the act as a preventive measure.

¹ Smith, A. (1776).

² Majaski, C. (2025).

³ Hall, M. (2023)

⁴ FRBSL. (2025, October 24)

⁵ McSweeney, T. (2017)

⁶ McSweeney, T. (2017)

⁷ Kropff, A. (2016)

⁸ LII. (2025)

⁹ Clark, O. L. (1948).

Sherman and others originally intended to protect consumers; however, their initial attempt resulted in unintended consequences. Soon after its inception, corporations leveraged the vague wording of ‘restraint of trade or commerce’ to prosecute unions. In fact, “its literal application would condemn all the concerted activities of striking, picketing and primary and secondary boycott where interstate commerce is restrained.”¹⁰ Corporations argued that any exceptions to the rule would have been explicitly noted in the original legislation; the Supreme Court upheld their interpretation in 1893. In 1908, the landmark Danbury Hatters case *Loewe v. Lawlor*¹¹ used precedent to hold a peaceful interstate boycott liable for damages. This case represented the broadest anti-labor interpretation of the law and marked the first recognition of union employment as economically insignificant.

B. The Clayton Antitrust Act

Modern academics recognize the positive effects of union employment¹²; at the time, however, corporations dominated the narrative by repeatedly invoking the act. Accordingly, Congress attempted to clarify the original intent. In 1914, Congress passed the Clayton Antitrust Act¹³ which explicitly excluded peaceful union activity from antitrust litigation. In essence, the act declared that, “the anti-trust laws shall not be construed to forbid the normal functioning of labor unions to lawfully carry out their legitimate objects.”¹⁴ The Clayton Act demonstrated the federal government’s commitment to protecting labor; however, Congress’s second attempt produced additional unintended consequences.

¹⁰ Clark, O. L. (1948).

¹¹ *Loewe v. Lawlor* | 208 U.S. 274 (1908).

¹² Banerjee, A. (2021).

¹³ Clark, O. L. (1948).

¹⁴ Clark, O. L. (1948).

The Supreme Court’s decision in *Duplex Printing Press Co. v. Deering*¹⁵ disappointed proponents of the Clayton Act. A majority of justices found that the act merely covered, ‘that which was permissible without it.’ According to the majority opinion, “there is nothing in the section to exempt such an organization or its members from accountability where it or they depart from its normal and legitimate objects and engage in an actual combination or conspiracy in restraint of trade.” Thus, labor unions continued to face corporate litigation and were held liable for the costs of collective bargaining. This precedent stained early antitrust efforts, punishing unions up until 1932¹⁶.

C. The Norris-LaGuardia Act of 1932

To avoid any further misappropriation of well-intentioned antitrust efforts, the Norris-LaGuardia Act expressly exempted peaceful pursuits of labor objectives.¹⁷ Its first implementation took place in *United States v. Hutcheson*,¹⁸ wherein two competing union leaders faced a jurisdictional dispute over a labor district. The affected union leader filed a criminal complaint using the act to formally indict and punish their competition.

Concern over union-on-union violence was short-lived, as the Supreme Court soon recognized the legal harmony established by the Sherman, Clayton, and Norris-LaGuardia antitrust acts that collectively protected the rights of laborers.

¹⁵ *Duplex Printing Press Co. v. Deering* | 254 U.S. 443 (1921)

¹⁶ *Bedford Cut Stone Co. v. Journeymen Stone Cutters’ Assn.*, 274 U. S. 37 (1927); *United Mine Workers of America v. Coronado Coal Co.*, 259 U. S. 344 (1922), later proceedings, 268 U. S. 205 (1925); *Aeolian Co. v. Fisher*, 40 F. (2d) 189 (C. C. A. 2d 1930) ; *Rockwood Corporation v. Bricklayers’ Local Union No. 1*, 33 F. (2d) 25 (C. C. A. 8th, 1929).

¹⁷ Clark, O. L. (1948).

¹⁸ *United States v. Hutcheson* | 312 U.S. 219 (1941)

According to the majority opinion in *United States v. Hutcheson*, “whether trade union conduct constitutes a violation of the Sherman Law is to be determined only by reading the Sherman Law and section 20 of the Clayton Act and the Norris-LaGuardia Act as a harmonizing text of outlawry of labor conduct.”¹⁹ This precedent constituted meaningful labor protection, despite flagrant and intentional union obstruction to interstate commerce.

III. Antitrust and Trusts

The United States has emphasized diversity, equity, and inclusion since its inception.²⁰ In fact, debates over systemic overrepresentation of rural areas in the senate preceded any consensus on the constitution. The founders initially intended to balance competing interests, preventing concentration of power in any specific branch, in order to properly represent the interest of constituents. Antitrust legislation presents another example of this storied tradition. Despite its initial flaws, the development of antitrust doctrine, along with union prevalence, ushered in an age of working-class empowerment.²¹ Lawmakers recognized the advantages of managers and capital owners and sought equal opportunity for Americans of all socioeconomic backgrounds.

Despite its significant limitations and unintended consequences, antitrust legislation was essential to the larger ‘trust-busting’ agenda. President Theodore Roosevelt, the most famous historical proponent of antitrust enforcement, used his presidential authority to shift policy towards trust-busting.²² According to the late president, “Great corporations exist only because they are created and safeguarded by

our institutions.” His administration argued that regulation preceded cooperation with corporations. President Roosevelt famously proclaimed, “The tremendous and highly complex industrial development which went on with ever accelerated rapidity during the latter half of the nineteenth century brings us face to face, at the beginning of the twentieth, with very serious social problems.”²³ Throughout his presidency, Roosevelt emphasized the importance of regulating the ‘costs of doing business.’ Consequently, his leadership led to the first major trust break-up in U.S. history.

A. *The Northern Securities Company*

By 1902, the Sherman Antitrust Act’s increasingly insidious application nearly delegitimized the legislation. President Roosevelt revitalized the antitrust movement, and brought a lawsuit to the Northern Securities Company; the company was labelled a ‘captain of industry’ due to its inordinate market share. The lawsuit was successful, and marked the first major breakup of a corporate titan. Over the next seven years, he would initiate 43 other lawsuits against market leaders.²⁴ He further cemented this commitment by establishing the Bureau of Corporations, which, though it lacked regulatory authority, represented a significant and permanent institutionalization of antitrust oversight within the federal bureaucracy.

B. *Anthracite Coal Strike*

The 1902 Anthracite Coal Strike marks the first noted occurrence of a president intervening in a labor-movement on behalf of laborers.²⁵ Due to his limited regulatory authority, President Roosevelt created dire circumstances for corporations that fostered

¹⁹ *United States v. Hutcheson* | 312 U.S. 219 (1941)

²⁰ Eddie, M. (2025).

²¹ Clark, O. L. (1948).

²² Milkis, S. (2023, August 28).

²³ Milkis, S. (2023, August 28).

²⁴ Milkis, S. (2023, August 28).

²⁵ Milkis, S. (2023, August 28).

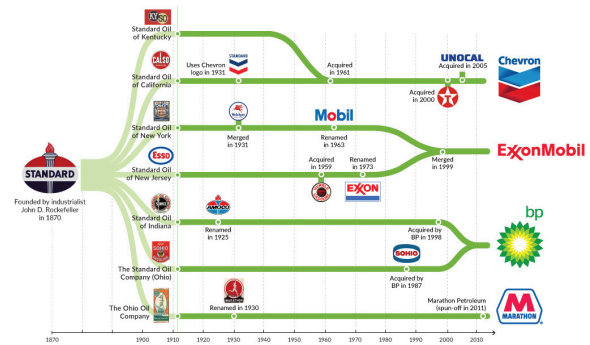
unhealthy work environments. Cooper notes: “He also talked about calling in the army to run the mines, and he got Wall Street investment houses to threaten to withhold credit to the coal companies and dump their stocks.²⁶” This corporate chokehold forced the coal mine to negotiate with the union, which resulted in a landmark victory for laborers.

President Roosevelt’s consistent commitment to consumers and laborers proved popular, and in his second term he established regulatory agencies to continue his efforts for years to come. The combination of investigative journalists or ‘muckrakers’ and regulatory agencies revealed and regulated the dreadful factory conditions of the early twentieth century.

C. Standard Oil

President Roosevelt had effectively breathed new life into the antitrust agenda, a momentum that culminated near the end of his second term with a case against the largest conglomerate the government had yet faced: Standard Oil. The 1911 court-ordered breakup of the company, three years after Roosevelt's administration initiated the suit, sent shockwaves through America's burgeoning corporate culture.²⁷ This landmark victory was aided by the work of muckraker Ida Tarbell, whose meticulous reporting had previously revealed the extent of Standard Oil's malfeasance to the public, inciting such outrage that the state of Kansas had symbolically "declared war" on the company as early as 1905.²⁸ While the theoretical economic role of monopolies was debated, muckrakers like Tarbell exposed their profound social costs and corrupt business practices. On May 18th, 1911, the Supreme Court ordered the Standard Oil

trust dissolved within six months. With its dominant market share disrupted, the original entity was broken into 34 new companies,²⁹ many of which have since consolidated into the modern-day energy giants Chevron, ExxonMobil, and Marathon Petroleum.



D. American Tobacco and Price Fixing

In 1907, a subsidiary of the American Tobacco Company was indicted for price-fixing in New York. This filing alerted the Justice Department to its predatory pricing models. President Roosevelt once again initiated a suit against the industry leader. Four years later, the Supreme Court recognized the American Tobacco trust as a monopoly and ordered it to dissolve.³⁰ Before its dissolution in 1911, the American Tobacco conglomerate had captured 85% of the cigarette market.

By selling cigarettes at or below cost, American Tobacco used its existing market share to price out competition. While price competition is generally seen as worse for businesses than for consumers, the trust's goal was market dominance rather than solvency. With time, the trust would have become an unshakeable facet of American life – that is, unless regulators broke up its consolidated power. Without regulation, American Tobacco could have become a market unto itself. Once a true monopoly is

²⁶ Milkis, S. (2023, August 28).

²⁷ Williams, K. (2019, November 5).

²⁸ Williams, K. (2019, November 5).

²⁹ Williams, K. (2019, November 5).

³⁰ Lee, J. (2023).

established, consumers have no choice but to accept any price a company may set.

The sixteen companies that emerged from the court-mandated dissolution of the trust held a dominant market share for a considerable time. The pre-existing connections and integrated operations among them provided a robust infrastructure that benefited all of the resulting business models.³¹

The breakup of American Tobacco established a legal precedent whereby monopolies were thereafter considered ‘unreasonable’ restraints of trade. This principle would soon prove essential as the federal government began to recognize the necessity of market regulation and consumer protection.

IV: Tools of the Trade

A. Anticompetitive Business Models

The FTC currently categorizes anticompetitive business structures into two essential types: vertical and horizontal integration. These two models focus on market share but do so in distinctly different ways.

Vertically integrated³² companies focus on supply chain specialization. Investing in every aspect of a supply chain gives a company a significant advantage over any competitor reliant on outside suppliers. By controlling all sections of a supply chain, a firm can sell itself all supplies at or below cost. Furthermore, it can use profits from more solvent sections of its business to finance those losses. Over time, it can use its extremely advantageous model to secure dominant market share by pricing competitors out. After securing enough market share, a company can set a universal price of a good. Without regulation, this model inevitably leaves

consumers with one choice and one price for a given market or product.

Horizontally integrated companies,³³ however, focus on acquiring competitors. Rather than prioritizing supply-chain efficiency, this model aims to purchase all existing competition, a strategy typically financed by capital from previous market dominance. Acquiring rival firms eventually gives consumers the illusion of choice while successfully consolidating intellectual, labor, and material capital within a single trust.

Whereas Standard Oil³⁴ utilized both vertical and horizontal integration, American Tobacco was unique for its specialized reliance on the vertical model. By selling cigarettes at or below cost, it effectively differentiated its market position based on price. This technique, known as price differentiation, allows a company to segment the market and dominate multiple price points. Even without explicit collusion, a company with unreasonably dominant market share can eventually dictate and effectively fix prices. While price fixing is often the final outcome, the more pressing question for regulators is whether the current legal definitions of anticompetitive business models remain satisfactory for identifying this threat.

B. Symptoms of Dominant Marketshare

In addition to defining anticompetitive business models, the FTC maintains a list of explicitly anticompetitive practices. Practices are distinguished from models based on their direct consumer impact and unethical nature.³⁵ This distinction highlights a central difficulty in antitrust enforcement: these tools and models are often the most optimal and efficient business strategies available. It logically follows, therefore, that

³¹ Carter, R. (2006).

³² Nguyen, S. T. (2022, March 5).

³³ Nguyen, S. T. (2022, March 5).

³⁴ Milkis, S. (2023, August 28).

³⁵ Nguyen, S. T. (2022, March 5).

corporations would learn to adopt them, forcing regulators to confront the challenge of restricting rational, yet harmful, economic behavior.

Tying Arrangements³⁶: It is illegal to force consumers to make a secondary transaction in order to buy a product. If Apple directly forced consumers to make a second purchase of a charger, the FTC would immediately intervene.

Agricultural representatives have recently filed a class-action lawsuit against the industrial farm equipment manufacturer John Deere. The suit alleges that the company “artificially restricted the market for authorized repair through a policy of forcing small, independent John Deere dealerships to consolidate or lose their franchise.”³⁷ Proving a company is engaging in such practices is notoriously difficult unless its actions fit the legal definition precisely. In this agricultural case, John Deere’s use of restricted repair software has artificially inflated maintenance costs for farm vehicles. Because farmers cannot obtain complex repairs from independent shops, they are effectively forced to accept any secondary terms and pricing that John Deere imposes. As the complaint states, “Deere’s tying arrangement between Deere Tractors and Repair Services had both the intent and effect of harming competition in the market for Deere Repair Services.”

Division of Markets³⁸: A key tenet of antitrust law is that while companies may naturally differentiate themselves by price or location, they cannot collude to divide markets. The presence of collusion is the defining element of an illegal scheme.

This illegal division of markets is a significant concern within the American healthcare system. The relationships between insurers, providers, and hospitals

are notoriously complex, and the effects of consolidation are often too obfuscated for consumers to discern, particularly during times of need. According to the KFF, “One or two health systems controlled the entire market for inpatient hospital care in nearly half of metropolitan areas in 2022.”³⁹ While health insurance companies are required to communicate regarding coverage to meet federal standards,⁴⁰ they are not forced to segment markets based on ‘networks.’ Such corporate consolidation is a top priority for the FTC, largely due to this market-leader control over essential inpatient services. These exclusive networks increase insurance profit margins by limiting consumer choice and restricting total benefits paid. Due to the intricate and relationship-dependent structures that take years to form, insurance companies effectively obscure divided markets from both regulators and consumers.

Bid Rigging⁴¹: For health insurance companies and industrial manufacturers alike, wholesale contracts are undoubtedly essential in far reaching business models. A company like John Deere needs steel to manufacture its tractors. Coincidentally, their competition does as well. If John Deere uses its dominant market share to rig the price of a contract bid, it creates a false sense of competition for material contracts. This collusion can happen in a number of ways: whether through horizontal integration, shell companies, or simple collusion, bid rigging fixes a new price equilibrium for supply-end contracts.

Without regulation, this can lead to vertical integration as manufacturers lose their solvency and large conglomerates sweep up the remains. Additionally, while similar in behavior, this tool is differentiated from private equity models that focus on a

³⁶ Nguyen, S. T. (2022, March 5).

³⁷ Kanter, J. (2023).

³⁸ Nguyen, S. T. (2022, March 5).

³⁹ Jamie, G. (2025, August 9).

⁴⁰ Albert, T. (2025, January 8).

⁴¹ Nguyen, S. T. (2022, March 5).

diversified portfolio of shares rather than an overarching plan.

Exclusive dealing⁴²: It is explicitly illegal for a company to prevent customers from purchasing from its competitors. A prominent modern example is the distinctive green text bubble Apple uses for messages from non-Apple devices⁴³ – a feature competitors have consistently alleged constitutes a form of market alienation or exclusive dealing. This "othering" of a rival product can diminish consumer appreciation for an entire competing ecosystem. Such practices exemplify the profound nuance in antitrust law, as they are nearly impossible to distinguish from simply effective, if aggressive, business strategy.

The landmark case *Epic Games v. Apple*⁴⁴ demonstrates that these nuanced strategies carry significant legal risk. In a September 2021 ruling, the court found that Apple's prevention of users from accessing outside e-commerce platforms for in-app purchases constituted illegal exclusive dealing. Epic successfully argued that Apple's developer restrictions were distinctly anti-competitive, citing how Apple immediately punished any developer on its App Store who advertised alternative storefronts. Furthermore, Apple's requirement that all developers use its own payment processing – which contractually siphoned thirty percent of all sales – cemented its control over the ecosystem.

Price-Fixing⁴⁵: While some of the tools described priorly are indicators of dominant market share and anti-competitive behavior, price fixing is an inevitable consequence of it. As corporations consolidate, price fixing is unavoidable.

'Competition' in an empty room only leaves one price.

What ultimately ties price-fixing to other anticompetitive tools is the element of collusion.⁴⁶ The primary difficulty for regulators lies in proving this illicit intent. This is why the FTC carefully differentiates mergers and collaborations based on their stated goals and potential outcomes.⁴⁷ The Commission assesses whether a partnership aims to improve "one or more business functions, such as production, distribution, marketing, purchasing or R&D, and thereby benefit... consumers by expanding output, reducing price, or enhancing quality, service, or innovation." However, if the FTC determines that the true purpose of a collaboration or merger is to disguise an otherwise illegal activity – such as price-fixing or division of markets – it will be prosecuted as a per se violation of antitrust law.⁴⁸

C. What Happens After?

The list of anticompetitive tools is not exhaustive, and the lines between them are often blurred; for instance, a division of markets can easily function as de facto exclusive dealing. Participants in such schemes are treated as cartels, and the Department of Justice criminally prosecutes participants in "hard-core cartel agreements." However, a significant challenge in enforcement is that while corporations are considered legal persons, it is difficult to criminally indict specific individuals without extensive proof. Consequently, regulators primarily rely on the deterrent effect of threatening severe fines and potential corporate dissolution.

The high stakes of such enforcement are illustrated by the recent case against

⁴² Nguyen, S. T. (2022, March 5).

⁴³ Apple Support. (2025).

⁴⁴ *Epic games, inc.. V. Apple, Inc.*, no. 21-16506 (9th cir. 2023)

⁴⁵ Nguyen, S. T. (2022, March 5).

⁴⁶ Nguyen, S. T. (2022, March 5).

⁴⁷ Nguyen, S. T. (2022, March 5).

⁴⁸ Nguyen, S. T. (2022, March 5).

Meta.⁴⁹ At the time of writing, the FTC’s antitrust complaint was denied by the courts. Had the FTC succeeded in proving Meta was an illegal monopoly, the court could have ordered the company to divest Instagram and WhatsApp, fundamentally restructuring two of the world’s most prevalent social media networks.

V. Modern Price-fixing

The landscape of anticompetitive behavior is evolving. While traditional forms of explicit collusion remain a focus of antitrust regulation, the advent of sophisticated algorithms has fundamentally disrupted the legal and economic conception of collusion itself. As with any dynamic field, the mechanisms of market manipulation have grown increasingly innovative.⁵⁰

Historically, enforcement actions could be levied against firms once overt collusion was substantiated. However, algorithmic collusion now challenges this regulatory paradigm by obscuring liability and distributing agency to non-human actors that fall outside established legal frameworks.⁵¹ Consequently, regulators are constrained by the very statutory language designed to ensure corporate accountability – a critical gap that the FTC has explicitly acknowledged in its recent policy proposals.

A. Provide Legal Exculpation and Sign Everything (PLEASE)

Modern pricing algorithms aggregate and anonymize data from major retailers across key U.S. economic sectors, including groceries, housing, higher education, energy, and healthcare. Their value lies not only in their universal applicability but, more critically, in their capacity to consistently

maximize profits. By inputting confidential information into these purportedly secure systems, actors like landowners, agricultural conglomerates, and healthcare providers facilitate a form of data-driven coordination.

This practice transcends mere privacy concerns; it challenges established legal doctrines of liability and reasonableness. These algorithms are designed to identify profit-maximizing prices with optimal efficiency, therefore, they can discover and implement anticompetitive pricing strategies at a scale⁵² and speed previously unattainable.

Agri Stats: Agriculture has not been a premier growth sector in the U.S. since the era of King Cotton. Consequently, many farmers and processors have struggled to achieve profitability without resorting to explicit collusion or sophisticated price differentiation. Into this environment, Agri Stats offered a solution: participants merely had to submit confidential data to an algorithm that aggregated and distributed competitively sensitive information on price, cost, and output among rival meat processors. While the pursuit of profit is essential for shareholders, this specific pricing model generates significant negative externalities, raising consumer costs and consolidating market power in ways that directly impact the American public.

According to the U.S. Department of Justice, “during the alleged conspiracy, turkey processing margins increased 300% in just three years, Tyson Foods saw its operating margin grow from 1.6% in 2009 to nearly 12% by 2016, and wholesale pork prices rose over 50% in five years after nearly a decade of relative price stability.⁵³” The profit margin optimization enabled by this system is unprecedented in the agricultural sector. According to the

⁴⁹ Jamali, L. (2025, November 18).

⁵⁰ Hazel, D. (2024, June 21).

⁵¹ McSweeney, T. (2017).

⁵² Hazel, D. (2024, June 21).

⁵³ *Justice Department sues agri stats* - United States Department of Justice. (2025, February 6).

Department of Justice complaint, the pricing model provides participants with ongoing counsel on whether to create artificial shortages in order to inflate grocery prices and boost margins. The scale of this influence is critical: Agri Stats aggregates and leverages data encompassing over 90% of broiler chicken sales, 80% of pork sales, and 90% of turkey sales in the United States.⁵⁴

This market dominance over essential food products grants Agri Stats significant leverage over the material conditions of American consumers. Furthermore, the firm's alleged manipulation extends beyond product markets into labor markets. The 2023 antitrust lawsuit was preceded by complaints accusing the company of utilizing similar algorithmic models to suppress wages, calculating the minimum possible compensation it could offer workers.

RealPage: In 2024, the Department of Justice, in cooperation with the Attorneys General of North Carolina, filed a lawsuit against RealPage for violating Sections 1 and 2 of the Sherman Antitrust Act.⁵⁵

RealPage exemplifies a parallel case in the housing sector, where its algorithmic network aggregates sensitive data from rental properties. While landowners are subject to antitrust regulation, the use of anonymized pricing algorithms allows them to circumvent personal liability, ultimately externalizing the cost onto renters. The economic impact of this conduct is significant: in 2024, the White House Press Secretary⁵⁶ cited estimates that RealPage's model was responsible for inflating rental

costs by approximately \$3.8 billion, achieved through engineered supply shortages and coordinated price increases.

The reach of this system is extensive; RealPage and its subsidiary, Lease Rent Options (LRO), are utilized in approximately ten percent of all U.S. rental units.⁵⁷ Crucially, while the underlying technology could theoretically recommend price reductions to maximize long-term occupancy, evidence suggests the company actively discourages such outcomes. A Department of Justice filing alleges that "RealPage takes extensive measures to prevent such behavior," structuring its software to incentivize the automatic acceptance of its price recommendations. This is achieved by imposing "onerous" burdens on users who wish to deviate. As documented by White House reporters, "landlords who reject a price recommendation are required to provide a 'business' reason for each individual floorplan in a building," which creates a deliberate administrative barrier to undercutting the algorithm's coordinated prices.⁵⁸

Ultimately, the failure of regulatory agencies to keep pace with these algorithmic cartels is actively eroding the ability of Americans to secure affordable housing.

Multiplan: The healthcare sector provides a further illustration of this phenomenon through the case of MultiPlan.⁵⁹ Initially, New York State attempted to regulate out-of-network pricing through FAIR Health, a data analytics platform overseen by the attorney general designed to calculate equitable rates. After its contract lapsed, major insurers seized the opportunity to create MultiPlan, a purportedly independent

⁵⁴ *Justice Department sues agri stats* - United States Department of Justice. (2025, February 6).

⁵⁵ *Justice Department sues RealPage* - United States Department of Justice. (2025, February 6).

⁵⁶ National Archives and Records Administration. (2024).

⁵⁷ *Justice Department sues RealPage* - United States Department of Justice. (2025, February 6).

⁵⁸ *Justice Department sues RealPage* - United States Department of Justice. (2025, February 6).

⁵⁹ Albert, T. (2025, January 8).

entity that aggregates patient and provider data. According to the American Medical Association, this enabled a coordinated shift: “the nation’s biggest health insurance companies decided they would all pay the same rate for out-of-network visits.”

This collective adoption has effectively supplanted competitive pricing. Today, roughly 700 of the nation’s 1,100 insurers – including the 15 largest – use MultiPlan rather than independently determining payment rates.⁶⁰ This structure allows insurers to prioritize shareholder returns and profit margins over fair provider compensation or consumer cost containment. Consequently, the healthcare market demonstrates how algorithmic, data-driven coordination can dominate an essential human need, creating a pricing mechanism that is both anti-competitive and notoriously complex to challenge in court.

VI. Modern Solutions

Regulators are actively looking for new approaches to regulate trusts. The narrative around antitrust, however, has been negative since the Reagan presidency. Despite their reformist goals, antitrust agencies have lost authority and appeal, which has impacted both their respect and political capital.⁶¹

Lawyers are masters of trade who live by the rule that ‘changing the law’ is never the answer. For regulators, however, there is no indication that any legal jargon exists to properly regulate this field.⁶²

History is often a reliable guide for new solutions. Algorithmic price fixing, however, is a familiar problem with no definite solution. Regulatory agencies are lucky enough to see these companies in daylight, regulation beyond a ‘reasonable’ level could quickly force these agencies to hide from public view, which would only

weaken any future attempts to assign liability.

A. The Consumer

Modern consumers are subjected to market structures deliberately designed to maximize profit extraction through sophisticated, data-driven means. Smith’s classical economic theory suggests that social welfare can be an unintended but beneficial consequence of capitalist competition, guided as if by an “invisible hand.” The rise of private algorithmic platforms demonstrates a technical capacity for industrial self-regulation. However, this capability does not confer legitimacy, as it replaces public oversight with a system designed solely for profit maximization, a goal often at odds with consumer welfare.

Most concerning is the prevalence of algorithmic price-fixing within markets for essential human necessities. These systems disproportionately target sectors like food, shelter, and healthcare – goods positioned at the base of Maslow’s hierarchy of needs. In these inelastic markets, consumer agency is severely limited, often resulting in submission to non-competitive pricing and the direct transfer of wealth to capital owners.

Corporate consolidation is thereby stunting collective prosperity in exchange for shareholder value. If society is at peace with corporate consolidation, then the corporations will regulate themselves. Otherwise, consumers must take the public policy burden on themselves; they must advocate for a return to the American tradition of antitrust.

B. Regulators

While consumers have little agency in the modern marketplace, regulators have a tremendous opportunity to cause real change and gain back public appeal. The former chair of the FTC, Lina Khan,

⁶⁰ Albert, T. (2025, January 8).

⁶¹ Pertschuck, M. (1982).

⁶² McSweeney, T. (2017)

mentioned in an interview that she was shocked how many, “unused or underused, authorities were just on the books. Congress had passed laws, some as uncontroversial as ‘it should be illegal to say this product is made in America when it is not.’⁶³” The Reagan administration significantly slashed FTC budgets, and no president has brought that funding back, which leaves the FTC “punching above its weight.” In this modern market, two outcomes seem desirable for regulators.

A Return to Tradition: In the early days of the labor movement, presidential authority was instrumental in compelling corporate submission. The aggressive regulatory measures that dissolved the Northern Securities Company a century ago possess the same potential to dismantle modern algorithmic cartels. Lina Khan’s tenure at the FTC exemplified this principle, advocating for a dynamic and efficient bureaucracy that the Commission must now leverage. Under Khan’s leadership, corporations viewed FTC regulation as such a substantial threat that they invested in public smear campaigns against her.⁶⁴ This forceful opposition stands as compelling evidence of her efficacy. A federal government authentically committed to consumer and labor protections could significantly strengthen this vital yet historically under-resourced agency.

Changing the Law: The power of thought and argument is fundamentally constrained by the limits of language. The Federal Trade Commission exemplifies this struggle as it works to define the legally precise terminology needed to confront algorithmic price-fixing – a practice that is plainly anticompetitive yet linguistically novel. As former FTC Commissioner Terrell McSweeney has stated, “when Congress enacted the Sherman and Clayton Acts over

a century ago, the term ‘robot’ did not exist.” That the agency continues to win cases using such dated precedent is remarkable. Yet, these laws urgently require modernization, especially as complex data systems redefine the very act of collusion.

Operators of these algorithmic systems are aware that current political intent to bolster the FTC is limited – a reality underscored by the administration’s documented stance toward the agency and its former Chair⁶⁵. What remains uncertain is when the necessary legislative updates will materialize. Even modest reforms, such as formally recognizing non-human agents under vicarious liability, could empower the agency to combat contemporary threats effectively.

The Biden administration proved an outlier in its antitrust focus. While the issue remains peripheral in the public consciousness, there is hope that its urgency will be recognized once more, particularly as the consequences of algorithmic collusion become increasingly tangible for consumers.

VII. Whose Prosperity is it?

Antitrust legislation represented necessary change in the 20th-century Progressive Era. Since the Reagan administration, however, this history has been reduced to mythology.⁶⁶ Weakened regulatory agencies are limited by their funding and public appeal, yet corporations continue to develop new methods of consumer deception.

Price-fixing by algorithm is assuredly a novel version of a cartel business model, though its underlying anticompetitive nature is no more creative than earlier forms. Policymakers must remember this history—the public advocacy, the labor-focused policies, and the commitment to protecting constituent

⁶³ Khan, L. (2025).

⁶⁴ Dayen, D. (2023, June 23).

⁶⁵ Dayen, D. (2023, June 23).

⁶⁶ Pertschuck, M. (1982).

interests that once defined effective antitrust enforcement.

‘Efficient bureaucracy’ is not an oxymoron.